

Similar Systems that already exist

Natural deduction game: [Natural Deduction by felikso](#)

A game with levels and a sandbox mode to build natural deduction proofs.

Natural deduction proof editor: [Natural deduction proof editor and checker](#)

A browser-based proof checker for Fitch-style natural deduction. You can enter premises, conclusion, and build the proof.

Natural deduction proof checker: [Proof Checker](#)

A variant / adapted site that uses the Open Logic proof engine.

Learning support tool for natural deduction proofs:

https://www.doc.ic.ac.uk/pandora/?utm_source

A learning support tool focusing on guiding the construction of proofs.

A site involving many logic tools : [The Logic Machine: Interactive Logic Software for Logic Primer](#)

A suite of logic tools including a proof checker and hint features for natural deduction, propositional & predicate logic.

Domino on acid: Visual / puzzle game using domino tiles as a metaphor for inference rules.

Logiko: [logiko](#)

A browser puzzle game where you build Fitch-style proofs under time constraints

QED: [QED](#)

Intended as an “interactive text / game”: you click/drag to prove statements, with levels/sections where new inference rules unlock as you progress

In term's of making a game, what do all of these have in common/where do they fall short

- Most systems are learning tools, not real games.
- Most tools check correctness but don't help users understand why a step fails or how to improve.
- All players face the same proofs regardless of skill level. Beginners get overwhelmed; experts get bored.
- Most tools use plain text boxes or proof trees. Little visual engagement.
- Students often find natural deduction abstract. Most tools don't link logic to real life scenarios
- Few tools track how players improve or where they struggle.

Tool / Site	Interaction Style	Pedagogy	Visual Engagement	Game Loop	Hint Quality
Natural Deduction (felikso)	Levels	Medium	Low	Basic	Low
Natural Deduction Proof Editor	Editor	Medium	Low	None	Basic
Proof Checker (Open Logic)	Editor	High	Low	None	Medium
Pandora	Guided	High	Low	None	High
Logic Machine	Multi-tool	High	Low	None	Medium
Domino on Acid	Puzzle	Low	Medium	Light	Low
Logiko	Timed Puzzle	Medium	Low	Basic	Low
QED	Narrative	Medium	Low	Unlock	Medium

How could this help me differentiate my project

- Include levels with progressive difficulty and scoring systems. This adds motivation for the user to advance through the game. Potentially even unlockables?
- Multi-level feedback? Could I offer two different types of hints, one being minor and one being major?
- Include animations and sound to make it more engaging, potentially even create a storyline for the user to follow. Idea I had was to do something involving wizards or something where the user must solve proofs to cast spells.
- Add a results page at the end which outlines where the user was strong and where they could improve

Importation of LaTeX

As I am planning on using react, importing some kind of tool to encode mathematical notation into my game should work well. **Math Jax**.

The problem I run into then will be, how does a user enter answers, and how do I verify the correctness of these answers. I believe I found the solution for both.

Entering answers: I believe I will have the format of the game as a drag and drop. So have all possible characters that can be used as a keyboard almost on the screen where the user will drag and drop these keys in whatever order they believe to be correct.

Verifying answer: I will have all these keys as buttons in react, this means I can assign each button a name, then to verify correctness I can verify the order of the buttons and if they are correct in sequence.

Summary

Existing natural deduction teaching tools are correctness-first and UI-minimal. They provide structured guidance but lack visual engagement, dynamic animations, and motivational game mechanics. Error feedback is typically textual, not explanatory or visual. No surveyed tool combines natural deduction pedagogy with a level-based progression loop, visual metaphors for inference, and animated proof tree interactions.